Introduction to Robotics Lab
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Architecture

System Overview

The robotic system, as a whole, is a pick-and-place pipeline consisting of a 4-degree-of-freedom robotic arm that grabs objects with its end-effector, guided by overhead vision from a camera that helps it locate its target and plan its trajectory. It can be divided into three subsystems, each responsible for a different primitive function.

- **Perception:** A camera is mounted above the robot, providing RGB feed and depth information about the workspace.
- **Planning:** A microcontroller is responsible for controlling the servomotors of the robotic arm to achieve its desired configuration.
- **Control:** The servomotors translate the motor commands received from the computing body into actual physical motion, leveraging their circuitry.

Kinematic Chain

The figure 1 highlights the joints and links involved in the robot—there are 5 links (counting the ground link as well) and 4 joints in between.

This is an RRRR arm, as all the joints are revolute. It has 4 degrees of freedom, as stated previously, which can be verified using Gruebler's formula: $N = 5$, $J = 4$, $m = 6$, $f_i = 1$ for $i = 1, 2, 3, 4$ $c_i = 5$ for $i = 1, 2, 3, 4$

$$Dof = m * (N - 1 - J) + \sum_{i=1}^{J} f_i$$
$$= 6 * (5 - 1 - 4) + 4 * 1 = 4$$

Out of the 6 required for full control of the position and orientation in free space, only 4 can be controlled. This constrains the robot to only move in a circle, which is less than the total grid paper area, and the end-effector cannot be oriented along two of the three axes.

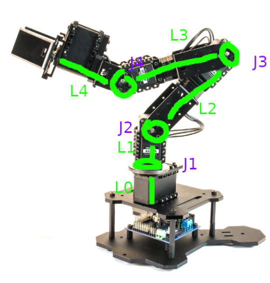


Figure 1.2: Phantom X Pincher Arm

Figure 1: Robotic arm with labeled joints and links

Sensors and Actuators

Sensors

The sensors involved in the system are:

- Intel RealSense Depth Camera
- Position sensors inside the servomotors (contributing indirectly towards sensing)

The primary purpose of the sensors in this system is to aid in perception and form the robot's world view, and the camera is primarily responsible for this. Other alternatives for this functionality can be Time of Flight-based cameras and sensors like LiDAR, which calculate depth using the round-trip time of light or ultrasonic waves to bounce back and return, offering great precision. But these are very expensive and suited for mobile robots with 3D taskspaces which require more elaborate sensing.

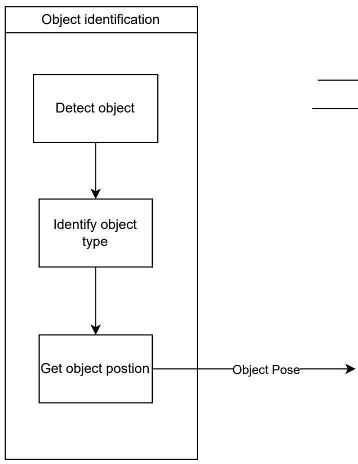
Actuators

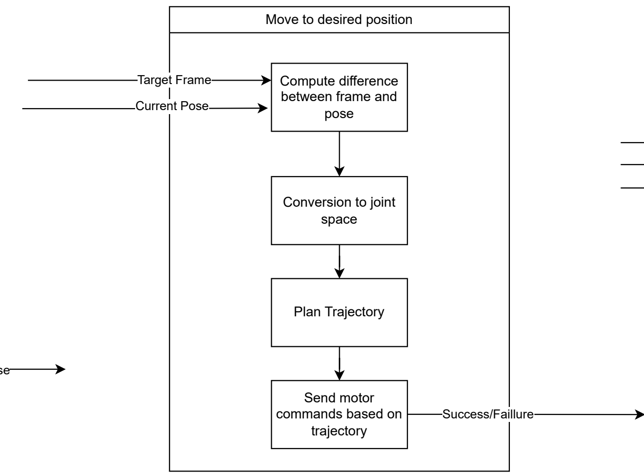
The actuators in the system are 5 Dynamixel AX-12A servomotors, which are responsible for translating the commands from the microcontroller (brain) to the physical movement of the arm at various joints.

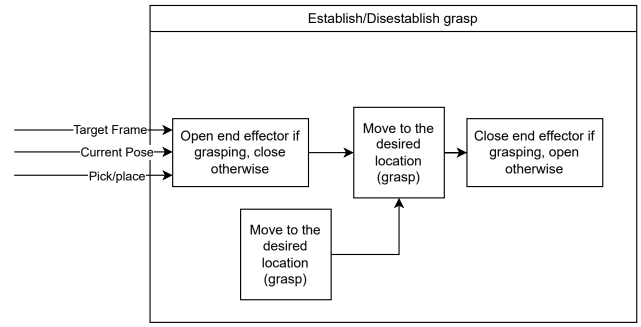
Inside the servomotor, there is a moving shaft and a potentiometer that work together as the slider of the latter is attached to the moving shaft in such a way that the mechanical movement of the shaft corresponds to the slider movement, adjusting the resistance and voltage along the path. This voltage is directly proportional to how much the shaft moved, allowing for precise position tracking and readjustment through feedback.

Data Flow

The data flow of the system is expressed by the following block diagram that describes the system as a composition of its different functions. The bottom flow describes the functionality of the system at a higher level, and the three blocks above it describe functions of object detection (via the camera), *move to desired position* as arm movement, and *end-effector grasping*, which are encapsulations over other repetitive and generic operations in the pipeline.


Object identification
Detect object
Identify object type
Get object postion
Object Pose


Move to desired position
Target Frame
Current Pose
Compute difference between frame and pose
Conversion to joint space
Plan Trajectory
Send motor commands based on trajectory
Success/Faillure


Establish/Disestablish grasp
Target Frame
Current Pose
Pick/place
Open end effector if grasping, close otherwise
Move to the desired location (grasp)
Close end effector if grasping, open otherwise
Move to the desired location (grasp)

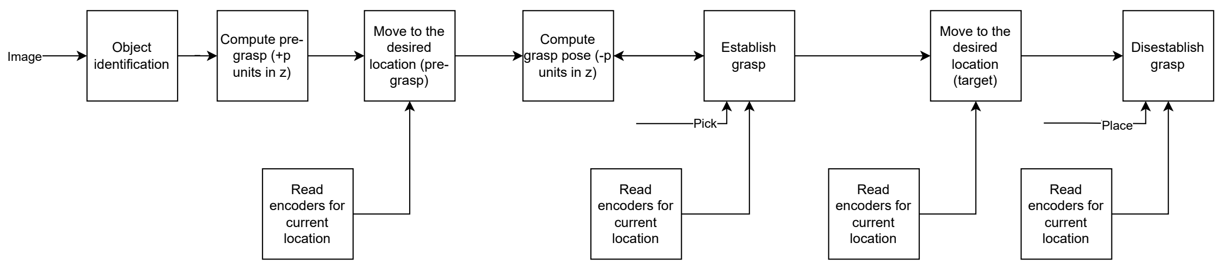


Image
Object identification
Compute pre-grasp (+p units in z)
Move to the desired location (pre-grasp)
Compute grasp pose (-p units in z)
Establish grasp
Move to the desired location (target)
Disestablish grasp
Pick
Place
Read encoders for current location
Read encoders for current location
Read encoders for current location
Read encoders for current location

Load-Carrying Capacity

For an estimate of the robot arm's load-carrying capacity, consider the equation 1 which describes the torque of the arm from the first link motor:

$$\tau = M_L g(L_1 + L_2 + L_3) + m_3 g(L_1 + L_2 + \frac{L_3}{2}) + m_3 g(L_1 + L_2) + m_2 g(L_1 + \frac{L_2}{2}) + m_2 g L_1 + m_1 g \frac{L_1}{2} \quad (1)$$

Given that the load on any motor should not exceed 1/5 of the stall torque, the equation can be arranged as follows:

$$M_L < \frac{\frac{1}{5}\tau_s - m_3(L_1 + L_2 + \frac{L_3}{2}) - m_3(L_1 + L_2) - m_2(L_1 + \frac{L_2}{2}) - m_2 L_1 - m_1 \frac{L_1}{2}}{L_1 + L_2 + L_3} \quad (2)$$

Substituting $L_1 = L_2 = L_3 = 105mm$ [1], $m_1 = 106.1g$, $m_2 = m_3 = 90.1g$, and $\tau_s = 1.5Nm$ [2] in equation 2, the value for the load capacity is obtained as:

$$\boxed{M_L < 246.6g}$$

This is close to the specified load capacity of 250g in the lab manual.

Bibliography

[1] "Phantomx pincher specifications," https://www.researchgate.net/publication/322222351_PhantomX_Pincher_Specifications, 2018, retrieved from ResearchGate.

[2] ROBOTIS Co., "How much is n·m when converted to kgf·cm?" https://en.robotis.com/model/board.php?bo_table=robotis_faq_en&wr_id=33&sca=GENERAL, n.d., accessed: 2026-01-27.